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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	BLUFFING
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG 4, SDG 8, SDG 9, SDG 10
Project / Solution Name	TELOS
Name (Leader)	Ivannov Tatang Kennedy
Email & Phone (Leader)	ivannovkennedy@gmail.com (+60177414536)
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	4 members (+1 leader)

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In the recent years, Artificial Intelligence has rapidly changed the way job market works, it creates new career opportunities while shifting the skills required for existing jobs. However, many fresh graduates, working adults, career switchers, and HR managers targeting on how to reskill their teams struggle to identify which skills they should develop to remain competitive in an AI-driven economy.

Looking at the data provided by *Hays Asia Salary Guide 2025*, 56% Malaysian professionals already use AI tools at work, yet only 26% have received formal training in the field. To top it all off, more than 600,000 workers will need reskilling within three to five years. These figures demonstrate the ever-growing gap between AI implementation and workforce readiness.

The problem matters because individuals may spend time and money learning skills that are becoming less relevant, while overlooking emerging career opportunities. Without accessible and up-to-date career guidance, these challenges may contribute to skills mismatches, reduced employability and economic inequality.

This issue is strongly targets Sustainable Development Goal (SDG) 4: Quality Education, particularly Target 4.4, which promotes relevant skills for employment and decent work. It also supports SDG 8: Decent Work and Economic Growth by encouraging workforce adaptability and productivity. Therefore, there is a need for an AI-powered solution that continuously tracks AI and job-market trends, identifies emerging and declining skills, matches individuals with suitable future careers and provides personalized reskilling pathways. Such a solution can help people make informed career decisions and prepare for future employment opportunities.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Telos is an AI powered web application designed to help university students, fresh graduates and working adults prepare for future job market changes. Derived from Ancient Greek, Telos is roughly translated to "ultimate aim". With this solution, we aim to provide personalized career recommendations by connecting users' existing skills and interests with emerging employment opportunities.

The prototype has five features: (1) a quick profile for users' skills, interests and available learning time; (2) a four-round crisis simulation with mission choices linked to real-world sustainability challenges; (3) a decision and skill report based on observable behaviors such as critical verification, calibrated AI trust, decision quality and adaptability; (4) explainable future-role matching that combines profile information, simulation evidence and occupational skill requirements; and (5) a future career pathway with moderate and rapid AI-adoption scenarios, ordered learning steps, estimated completion times, verified learning links and relevant SDG mappings.

ESCO and O*NET will provide occupational and skill information, while an updatable labour-market source such as OpenDOSM will add dated market signals. BGE-M3 embeddings will support semantic matching. A language model will evaluate free-text decisions using structured JSON outputs, while a predefined rubric and scenario-controlled events keep scoring more consistent. Supabase PostgreSQL will store profiles, scenario results and career data; Replit will support development and deployment, and RunPod will host AI inference.

TELOS aims to make career planning more engaging and evidence-informed. Its report will show behavioural examples and explain its limitations: one short simulation cannot determine a person's overall professional ability, and career recommendations are guidance rather than guarantees.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Telos will use the following technologies to develop a functional, deployed web prototype:

- **Frontend: Next.js/React**, developed in **Replit**, for the user profile form, crisis simulation and career dashboard.
- **Backend: FastAPI (Python),** served by **Uvicorn**, to manage game state, scenario logic, scoring and career matching.
- **AI Integration: Google Gemini 2.5 Flash API** for structured evaluation of free-text decisions and behavioural analysis. **Pydantic v2** validates AI responses against predefined JSON schemas.
- **Semantic Matching: BGE-M3 embeddings** via **RunPod** for matching users' skills with occupational requirements.
- **Database: Supabase PostgreSQL** with pgvector to store user profiles, simulation logs, skills, occupations, role matches and learning pathways.
- **External Data Sources: ESCO** and **O*NET** for occupational skills; **OpenDOSM Job Market Insights** for Malaysian labour-market data; and the **UN SDG API** for official SDG goals and targets.
- **Deployment: Render** to host the application services and support public access to the deployed prototype.

The end-to-end flow is: user profile → crisis simulation → structured behavioural evidence → skill report and role matching → personalised career pathway. The system will use synthetic profiles during testing and show data sources and dates for labour-market signals.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team will apply design thinking to the difficulty university students and fresh graduates face when identifying transferable skills and choosing career-development steps. Users will complete a quick skills profile, select an SDG-linked crisis mission, make decisions across four rounds, and receive a behavioural report with examples from their own responses.

To ensure reliability, we will use predefined scenarios and scoring rubrics, structured JSON outputs, and synthetic profiles to test behavioral reports, career matches, learning resources and SDG mappings. Career recommendations will combine users' stated skills, observed decisions and ESCO/O*NET occupational requirements.

To ensure long-term sustainability, ***Telos*** will use public datasets and periodically update job-market information instead of relying on fixed career lists. We will recommend free or low-cost learning resources to keep reskilling accessible, while using a modular system that allows new features and data sources to be added without rebuilding the application. We will also prioritise efficient resource usage and maintain transparent data sources to support reliable updates.

For Phase 2, we plan to integrate OpenDOSM job-market trends, provide verified learning pathways with estimated completion times, and introduce moderate and fast AI-disruption scenarios. Testing and user feedback will guide improvements to ensure the application remains practical, accessible and relevant as job-market needs change.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	*Primary SDG 4 (Quality Education), Target 4.4*: By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. *Supporting SDG 8 (Decent Work and Economic Growth), Target 8.5*: By 2030, achieve full and productive employment and decent work for all women and men, including young people and persons with disabilities. Reference: United Nations. (n.d.). Goal 4: Quality Education. https://sdgs.un.org/goals/goal4
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The World Economic Forum's Future of Jobs Report 2025 estimates that 41% of workers' core skills in Malaysia are expected to change by 2030, highlighting the growing need for career guidance and continuous skills development.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Skills-gap reduction rate (%): The percentage of identified skill gaps addressed through the personalized learning pathway, measured per user profile.

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve a 30% reduction in identified skills gaps among test profiles by the final prototype presentation, measured using the project's skills-matching and pathway assessment system. Reference: World Economic Forum. (2025). The Future of Jobs Report 2025. https://www.weforum.org/publications/the-future-of-jobs-report-2025/

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
ESCO Skills and Occupations	Dataset	CC BY 4.0
OpenDOSM Job Market Insights	Dataset	Public Domain (US government)
O*NET Database	Dataset	Malaysian government open data
BGE-M3 (BAAI)	AI Model	MIT License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24 October 2026